

## PART 1

What is React and why do we use it?

Ans: React is an open-source JavaScript library for building user interfaces (UI) and builds user interfaces out of individual components.

What is JSX? How is it different from HTML?

Ans: JSX (JavaScript XML) is a syntax extension for JavaScript that allows developers to write HTML-like markup directly within JavaScript code. HTML is a markup language which browser natively understands but JSX is a syntactical extension that integrates markup with JavaScript logic and requires compilation.

What is the purpose of the useState hook?

Ans: The useState hook in React is used to add state to functional components. Before hooks, only class components could have state. Now, with useState, functional components can store and manage dynamic data that affects what gets rendered.

Why do we use 'key' in lists?

Ans: In React, we use the key prop in lists to help React identify which items have changed, been added, or removed when the list re-renders. Without keys, React would have to re-render the entire list every time, which is less efficient and can cause bugs in component state.

What does useEffect do?

Ans: In React, the useEffect hook is used to perform side effects in functional components. Such as fetching data from an API, setting up timers and updating the document title.

What are controlled components in React?

Ans: In React, a controlled component is a form element (like <input>, <textarea>, <select>) whose value is controlled by React state. In other words, React is the "single source of truth" for the input's value, rather than the DOM managing it internally.

## PART 2

```
import React from "react";
```

```
const Todo = () => {
```

```
  const [todos, setTodos] = React.useState([]);
```

```
  const addTodo = () => {
```

```

// Create a new array instead of mutating state
setTodos(prevTodos => [
  ...prevTodos,
  `New Task ${prevTodos.length + 1}` // optional: add a unique label
]);
};

return (
  <div>
    <button onClick={addTodo}>Add</button>
    <ul>
      {todos.map((todo, index) => (
        <li key={index}>{todo}</li> // key added
      ))}
    </ul>
  </div>
);
};

```

export default Todo;

Part 3 – Practical Task → [github](#) → [practicaltask](#)